PRACTICAL REPORT

AWS Cloud & Data Engineering

Amazon S3 • AWS Glue • Linux • SQL

Submitted by

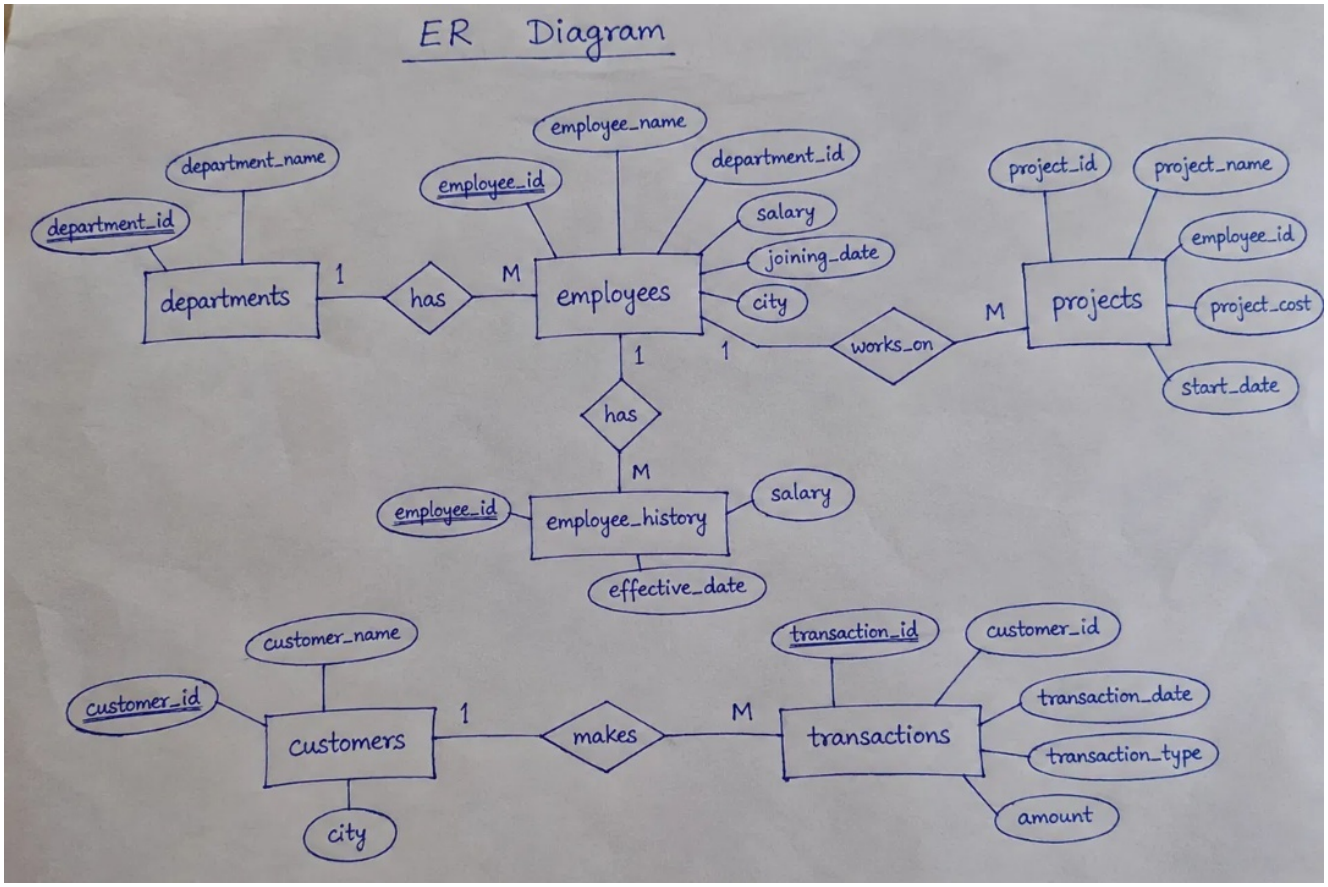
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Programme	B.E. — Computer Science and Engineering
Batch	2023–2027
Submission date	11 October 2026

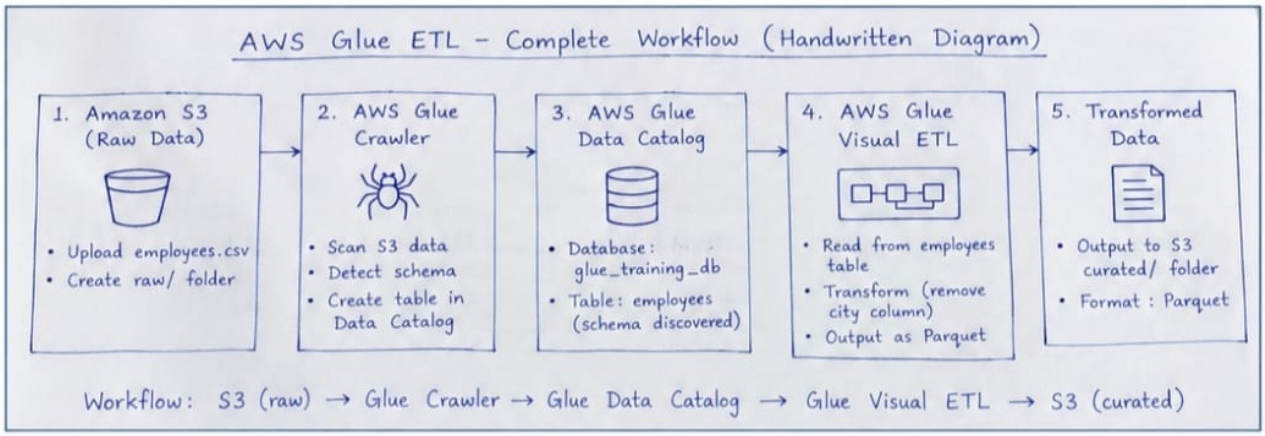
Department of Computer Science & Engineering

DIAGRAMS: ER DIAGRAM & AWS GLUE ETL WORKFLOW

ER Diagram



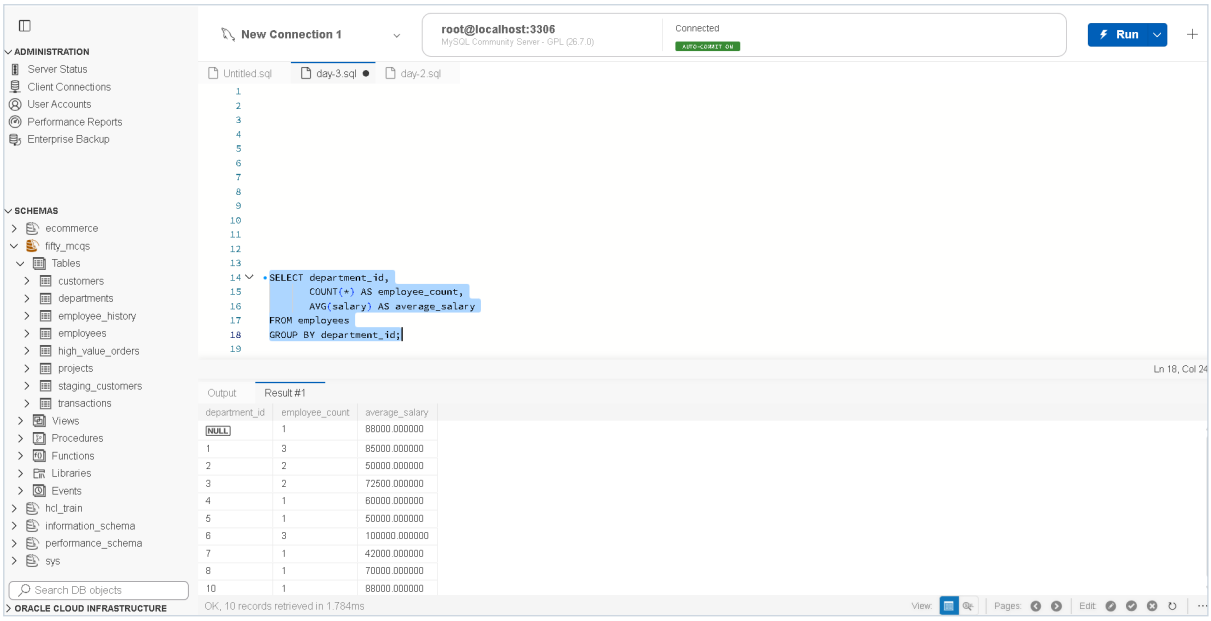
AWS Glue ETL — Complete Workflow



SQL Practice (MY SQL WORKBENCH)

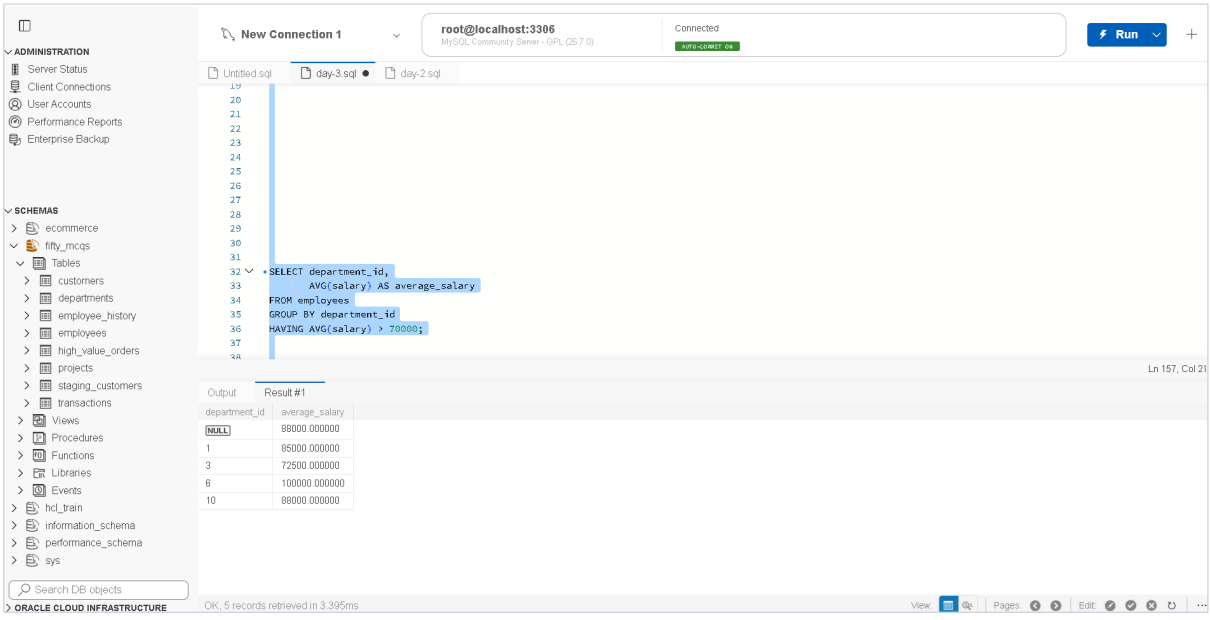
1. Department-Wise Employee Count and Average Salary

Purpose: Groups employees by department to show the headcount and average salary of each department.



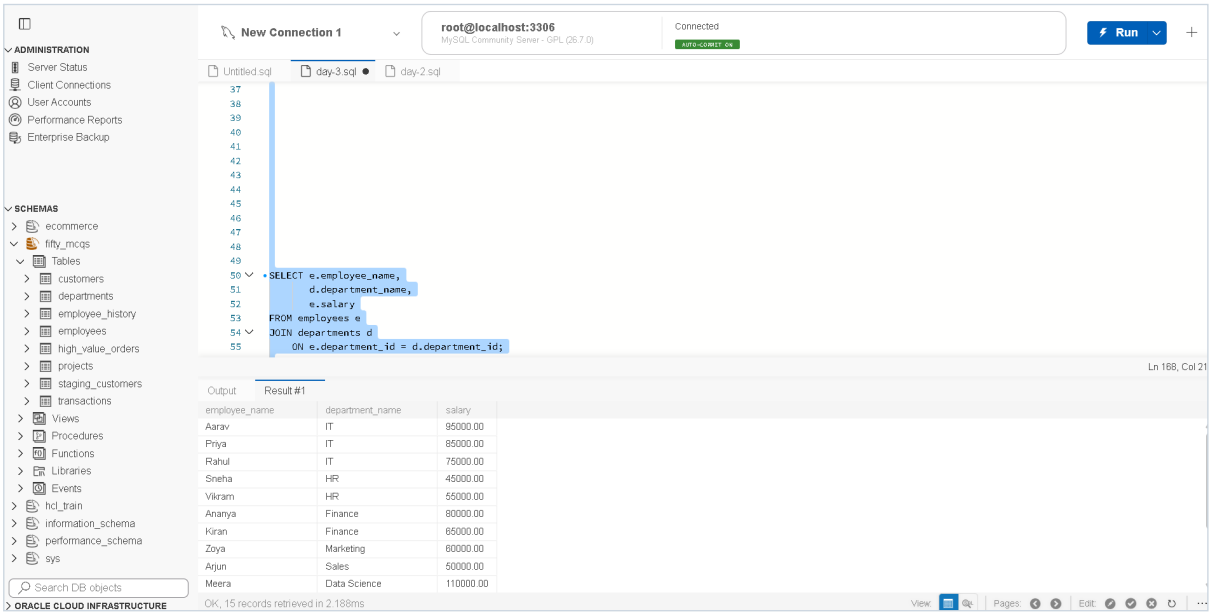
2. Departments With Average Salary Above 70,000

Purpose: Uses HAVING to keep only the departments whose average salary is greater than 70,000.



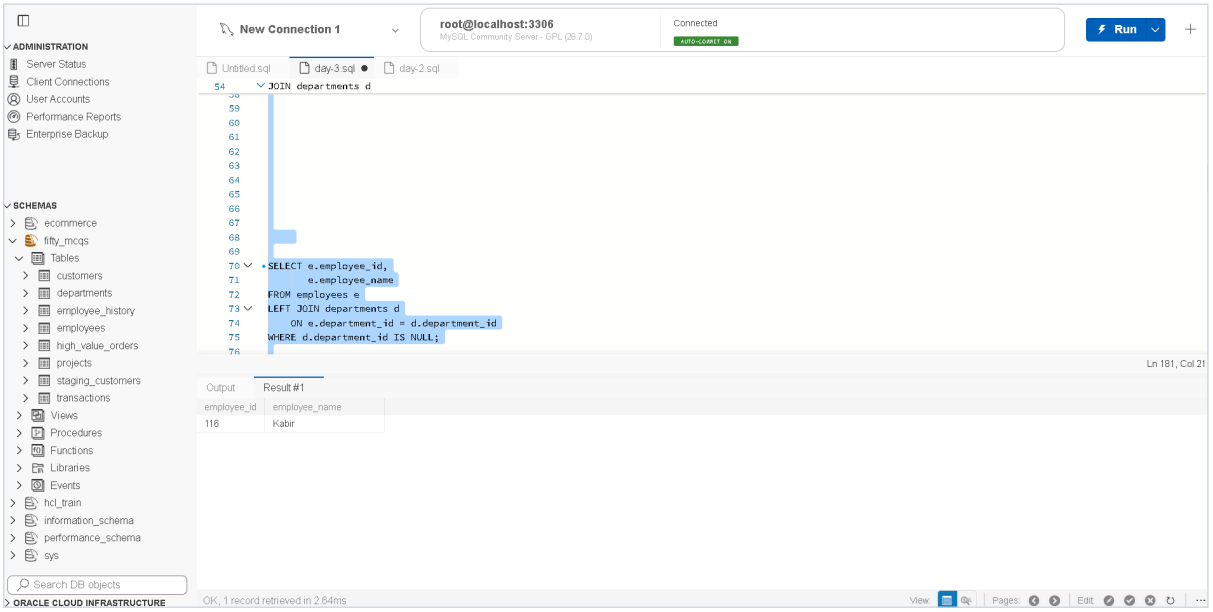
3. Employee Names With Department Names and Salaries

Purpose: Joins employees to departments to list each employee with the department name and salary.



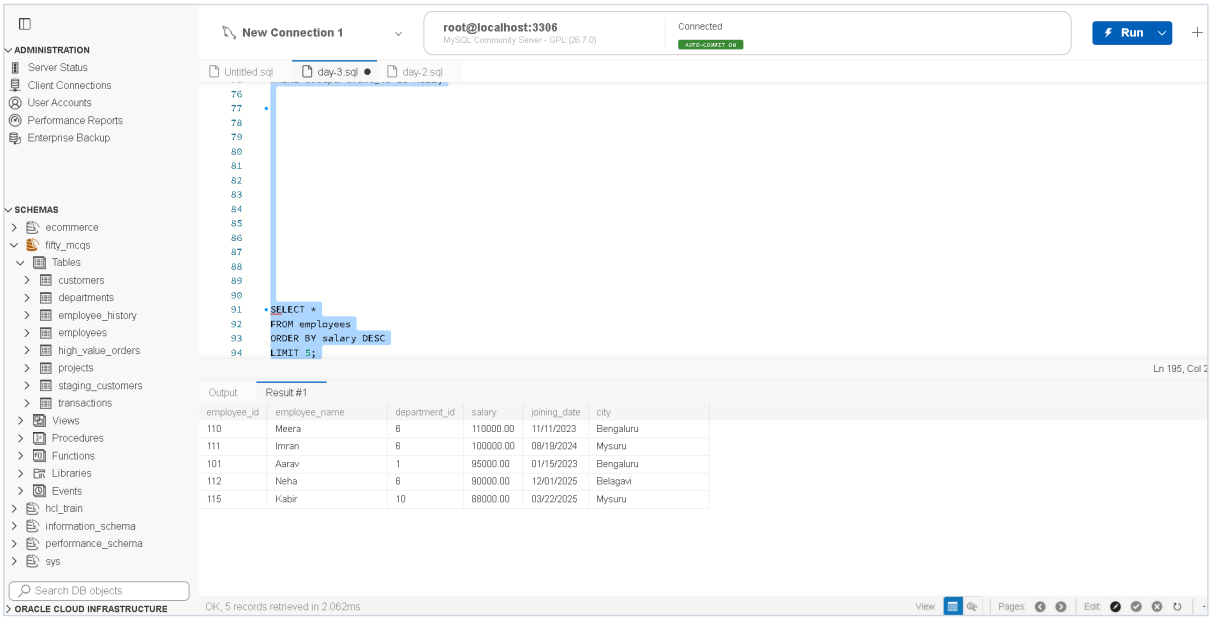
4. Employees Without a Matching Department

Purpose: Uses a LEFT JOIN to find employees whose department_id has no matching row in departments.



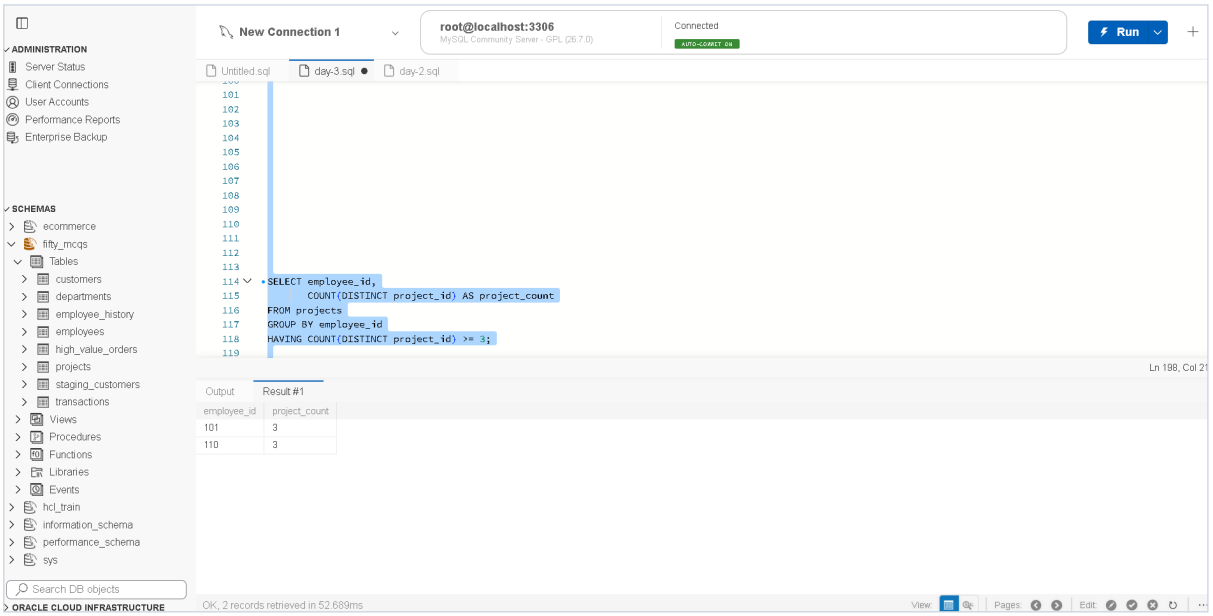
5. Top 5 Highest-Paid Employees

Purpose: Lists the five employees with the highest salaries.



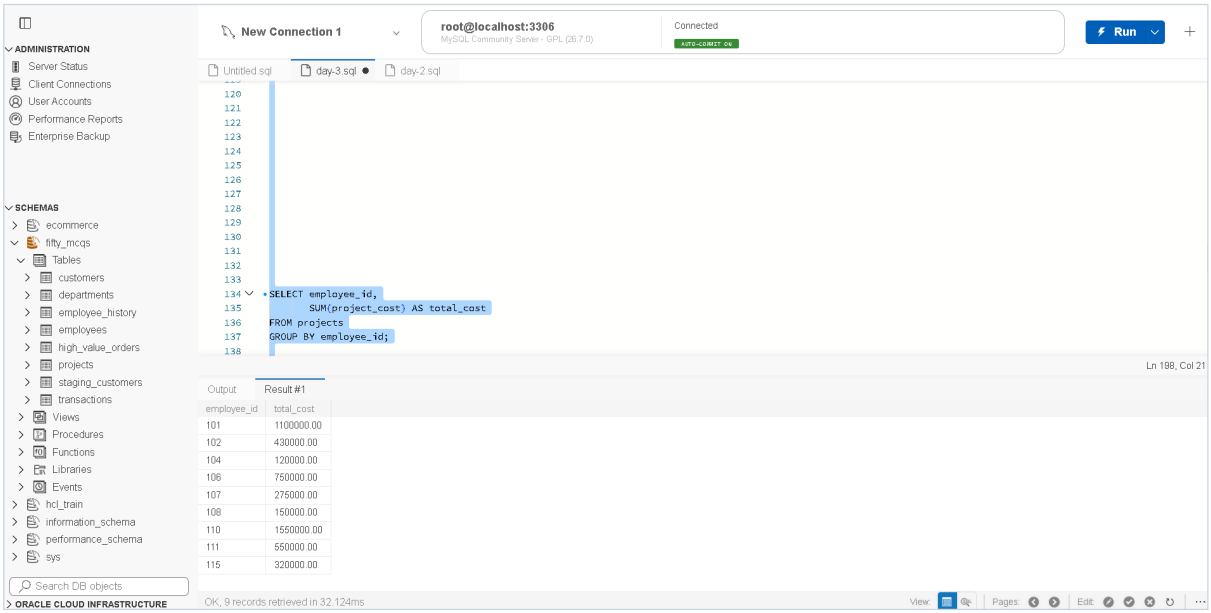
6. Employees Assigned to Three or More Projects

Purpose: Counts distinct projects per employee and keeps those with at least three projects.



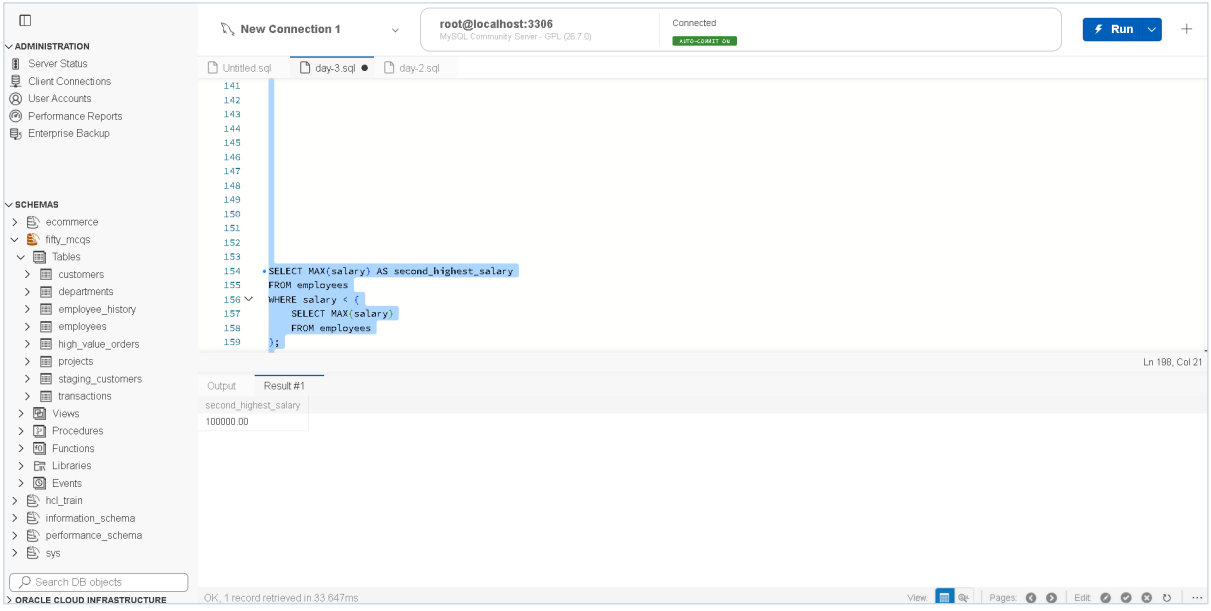
7. Total Project Cost per Employee

Purpose: Calculates the total project cost handled by each employee.



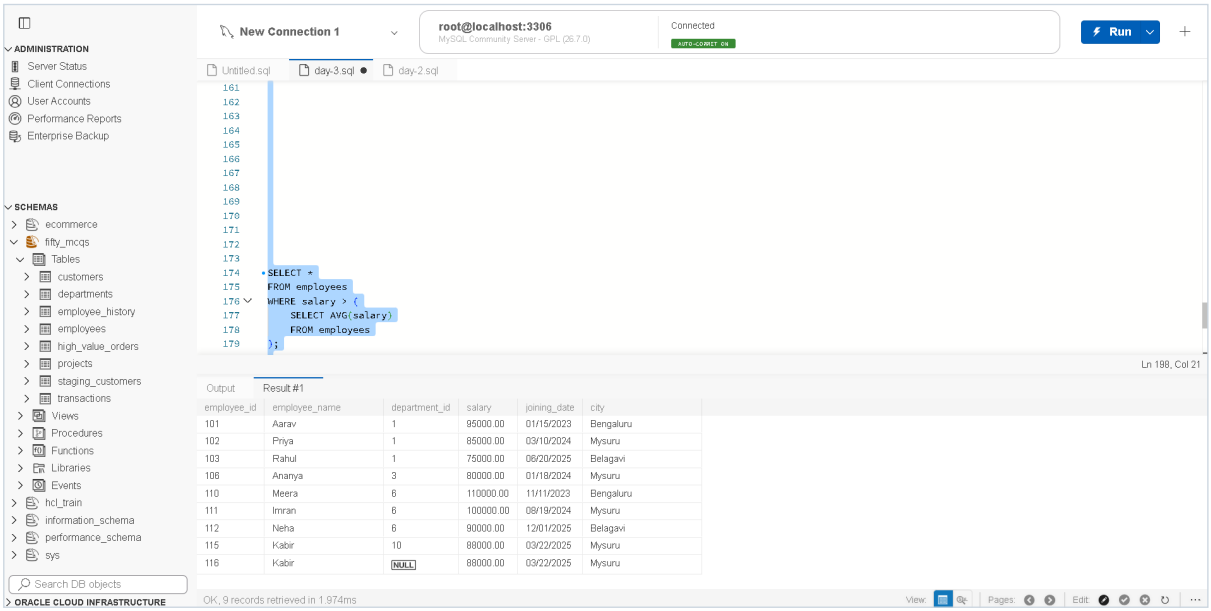
8. Second-Highest Salary

Purpose: Finds the second-highest salary by excluding the maximum salary with a subquery.



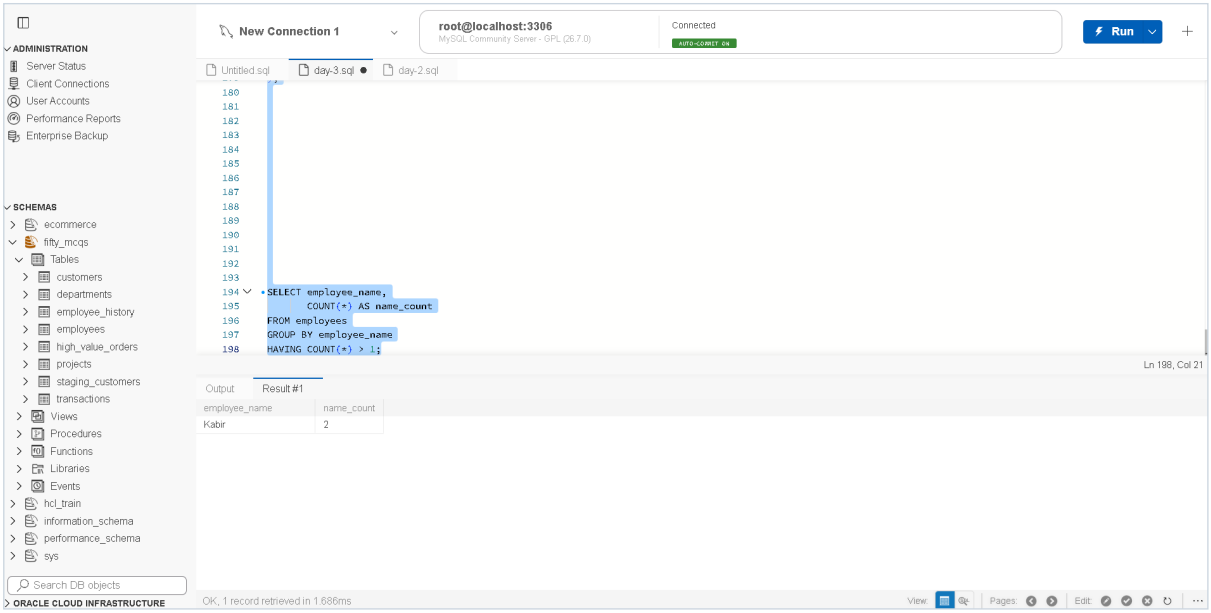
9. Employees Earning Above the Average Salary

Purpose: Lists employees whose salary is higher than the average salary of all employees.



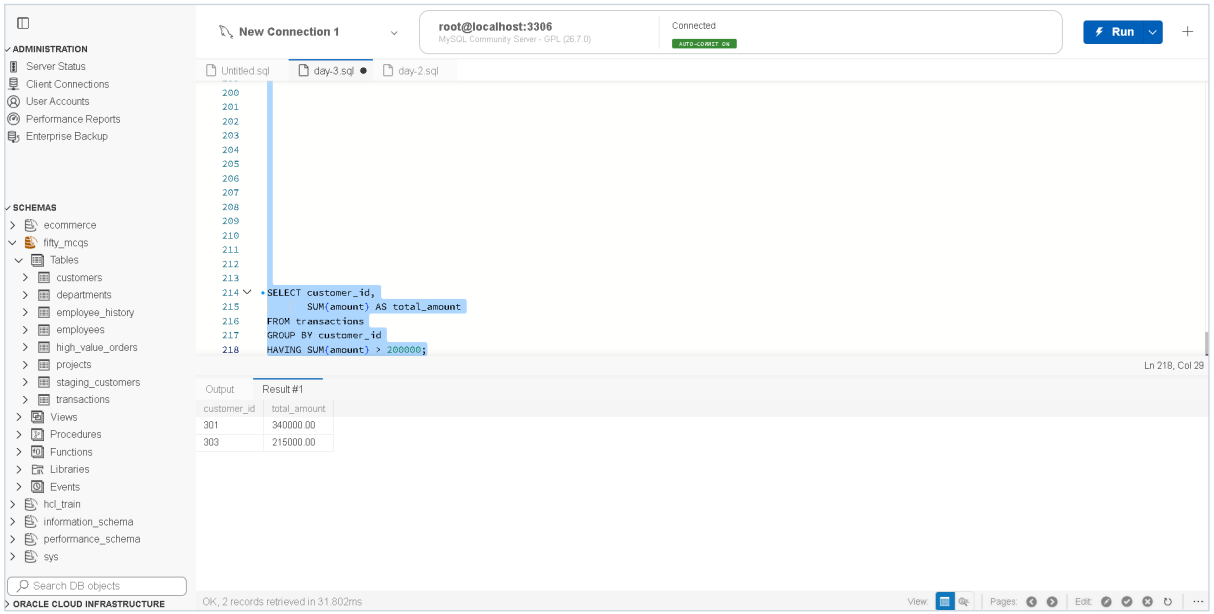
10. Duplicate Employee Names

Purpose: Finds employee names that appear more than once in the employees table.



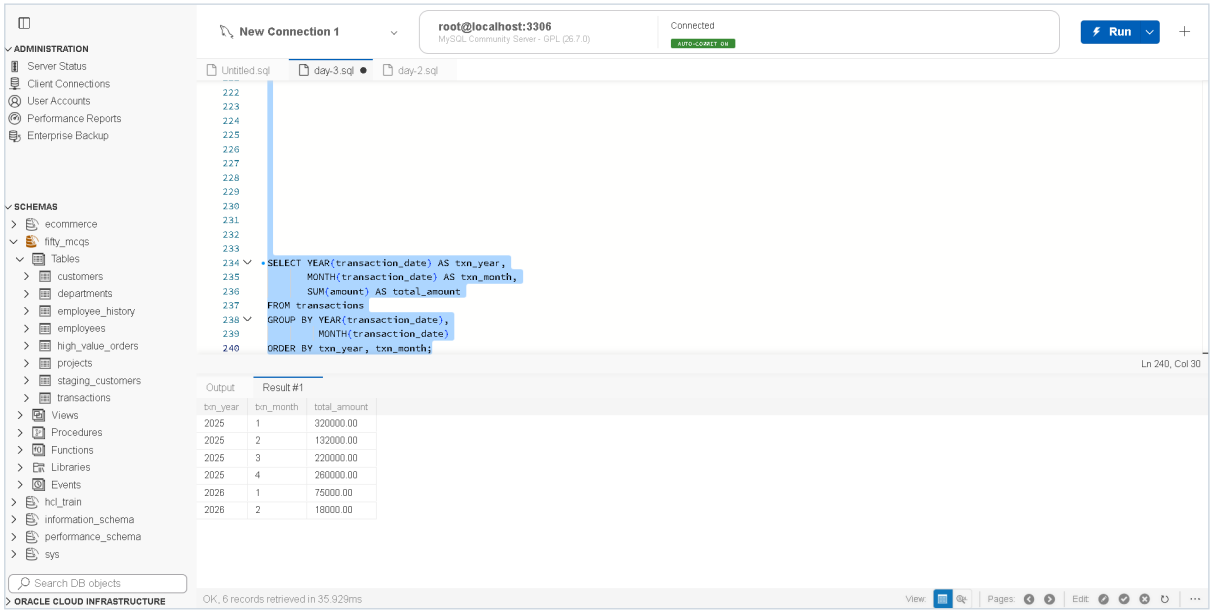
11. Customers With Total Transactions Above 200,000

Purpose: Sums the transaction amount of each customer and keeps those whose total exceeds 200,000.



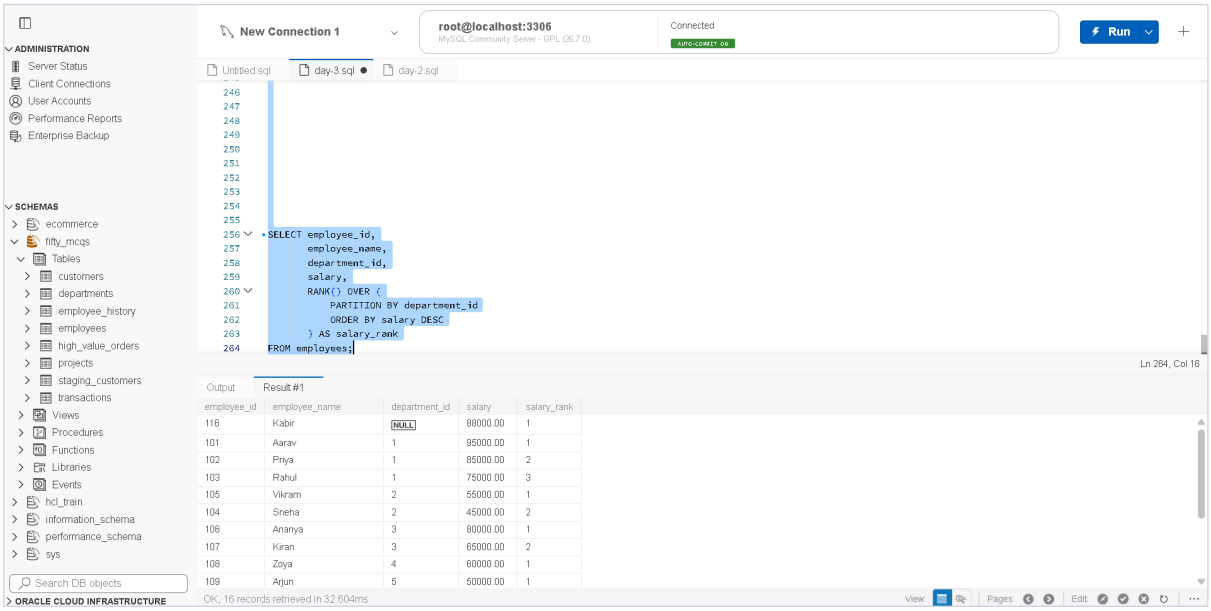
12. Monthly Transaction Totals by Year

Purpose: Summarises the total transaction amount for every year and month, in chronological order.



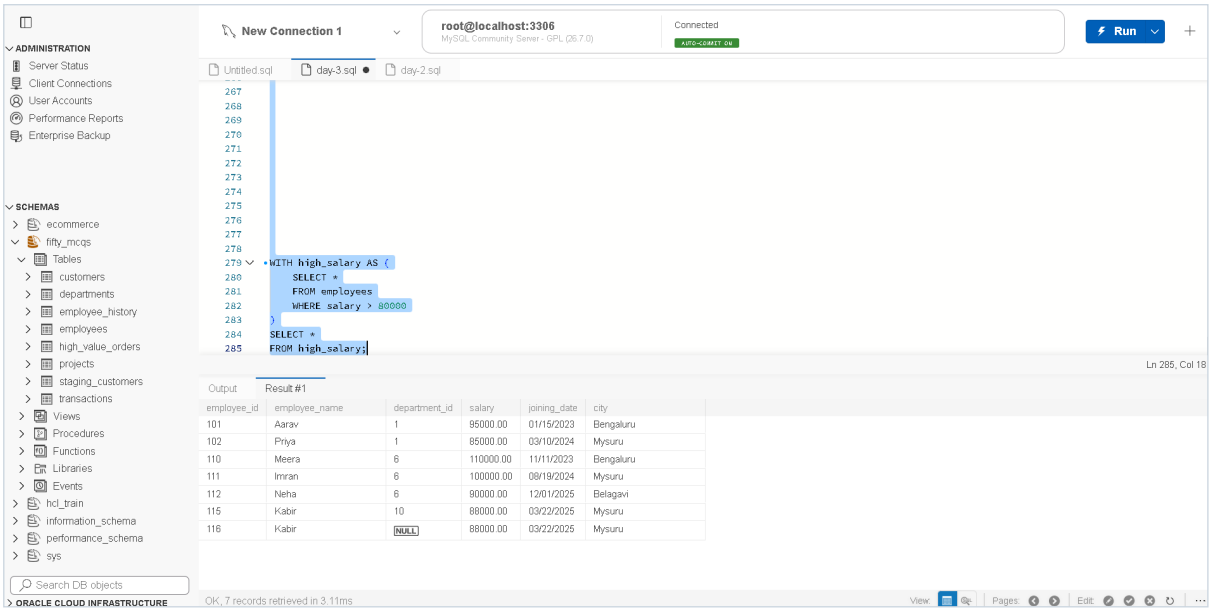
13. Salary Rank Within Each Department (RANK)

Purpose: Ranks employees by salary inside each department using RANK() with PARTITION BY.



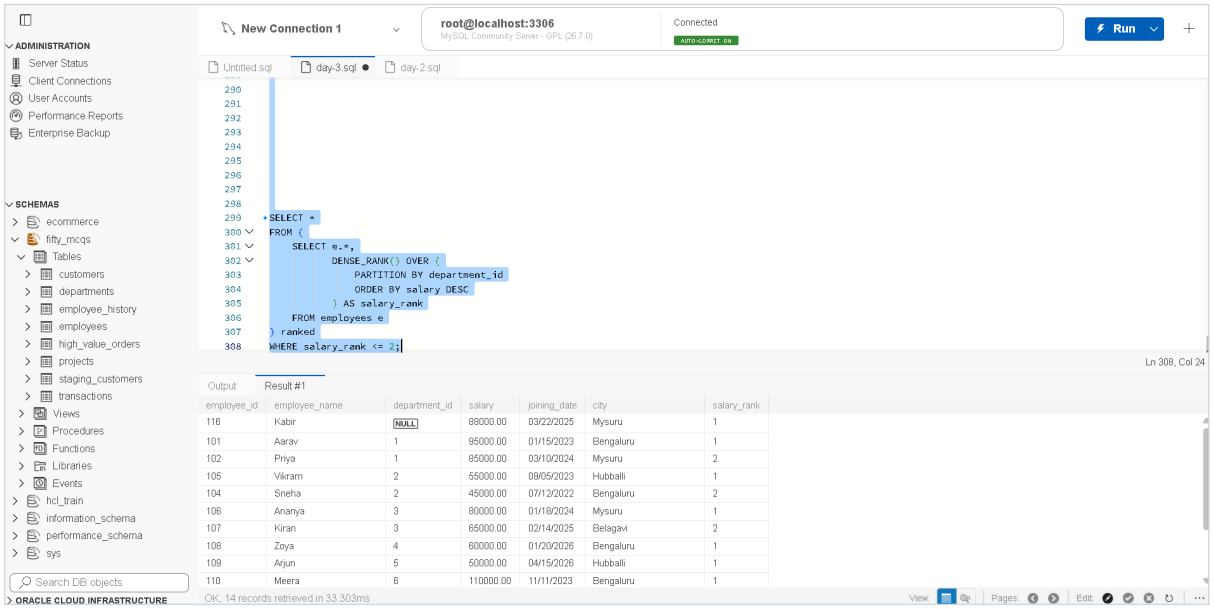
14. High-Salary Employees Using a CTE

Purpose: Defines a common table expression to select employees earning more than 80,000.



15. Top Two Salaries per Department (DENSE_RANK)

Purpose: Ranks salaries within each department using DENSE_RANK() and keeps ranks 1 and 2.



Linux Practice

1. Basic Linux commands

No.	Command	Use
1	pwd	Displays the current working directory.
2	whoami	Displays the current logged-in username.
3	date	Displays the current system date; this Windows prompt also asks for a new date.
4	uname -a	Displays system and kernel information (MSYS environment on this machine).
5	hostname	Displays the computer's host name.
6	ls -lah	Lists files, including hidden files, with details.
7	ls	Lists files and directories in the current location.
8	cd linux_practice	Changes the current directory to linux_practice.
9	cd ..	Moves to the parent directory.

2. Directory and file management

No.	Command	Use
10	mkdir linux_practice	Creates a directory named linux_practice.
11	touch notes.txt	Creates an empty file named notes.txt.
12	echo "Linux Practical" > notes.txt	Writes text to notes.txt, replacing existing content.
13	cat notes.txt	Displays the contents of notes.txt.
14	cp notes.txt backup.txt	Copies notes.txt to a new file named backup.txt.
15	mv backup.txt renamed.txt	Renames backup.txt to renamed.txt.
16	rm renamed.txt	Deletes the file renamed.txt.

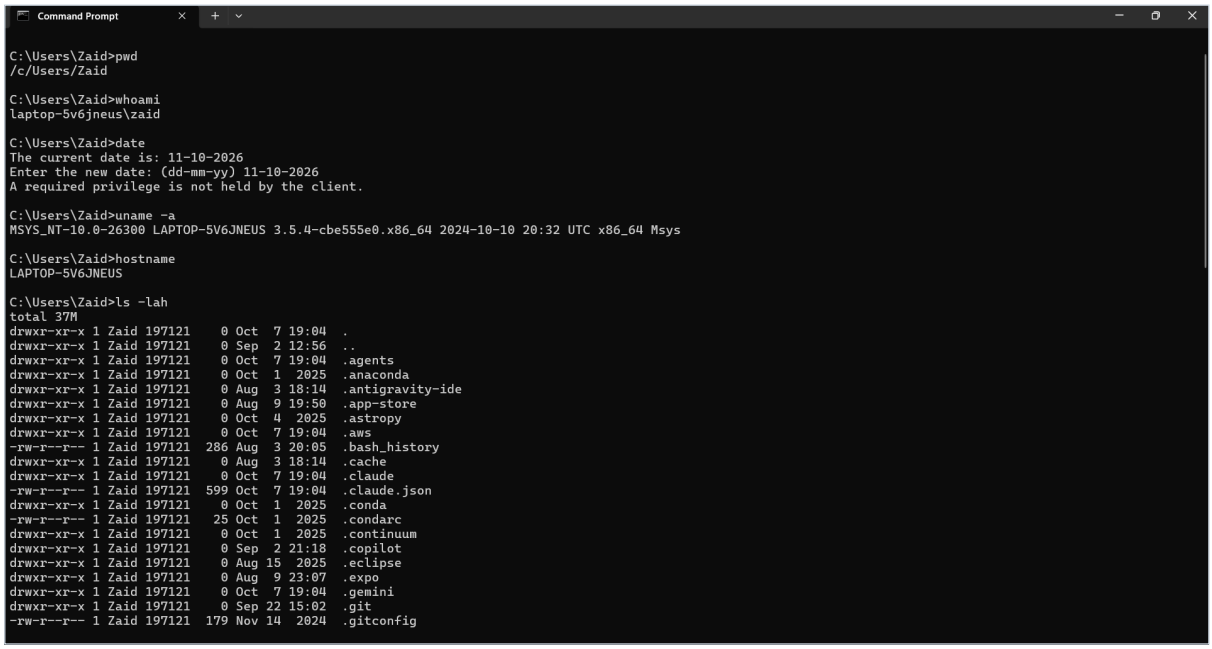
3. Permissions and disk usage

No.	Command	Use
17	chmod 644 notes.txt	Sets notes.txt to 644: the owner can read and write; group and others can only read.
18	df -h	Shows total, used and available disk space of each drive in human-readable units.
19	du -h	Shows the disk usage of directories and sub-directories in human-readable units.

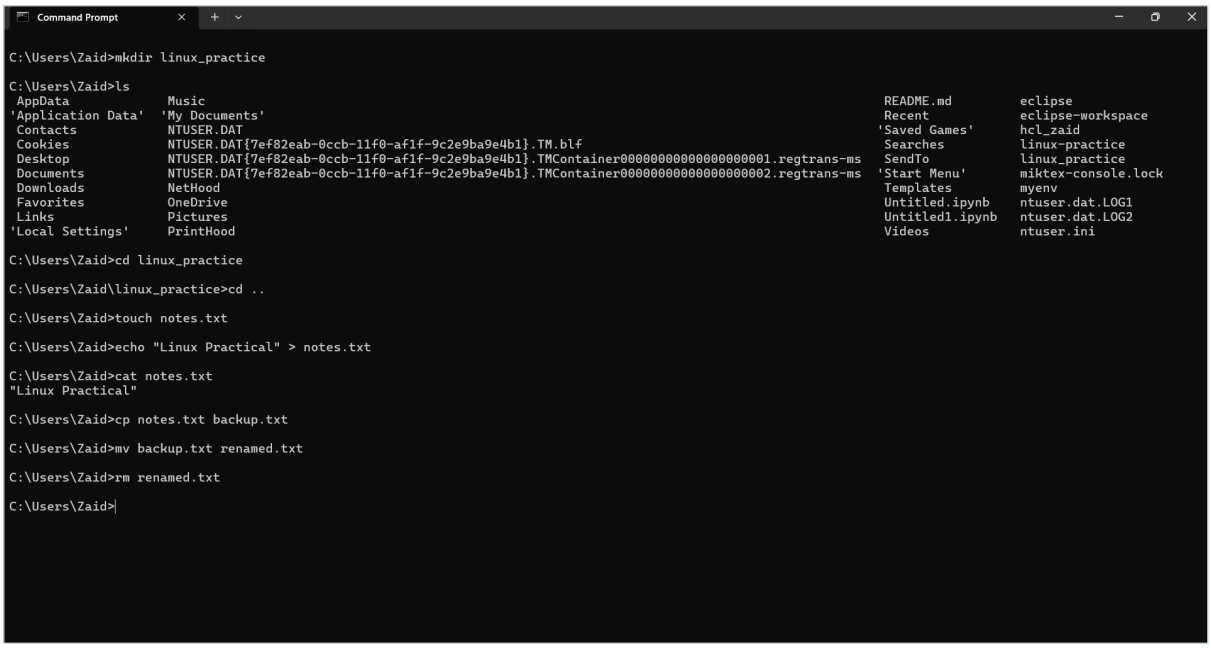
4. Terminal output

Commands were executed in a Windows Command Prompt using the Git/MSYS Unix utilities (see the uname -a output).

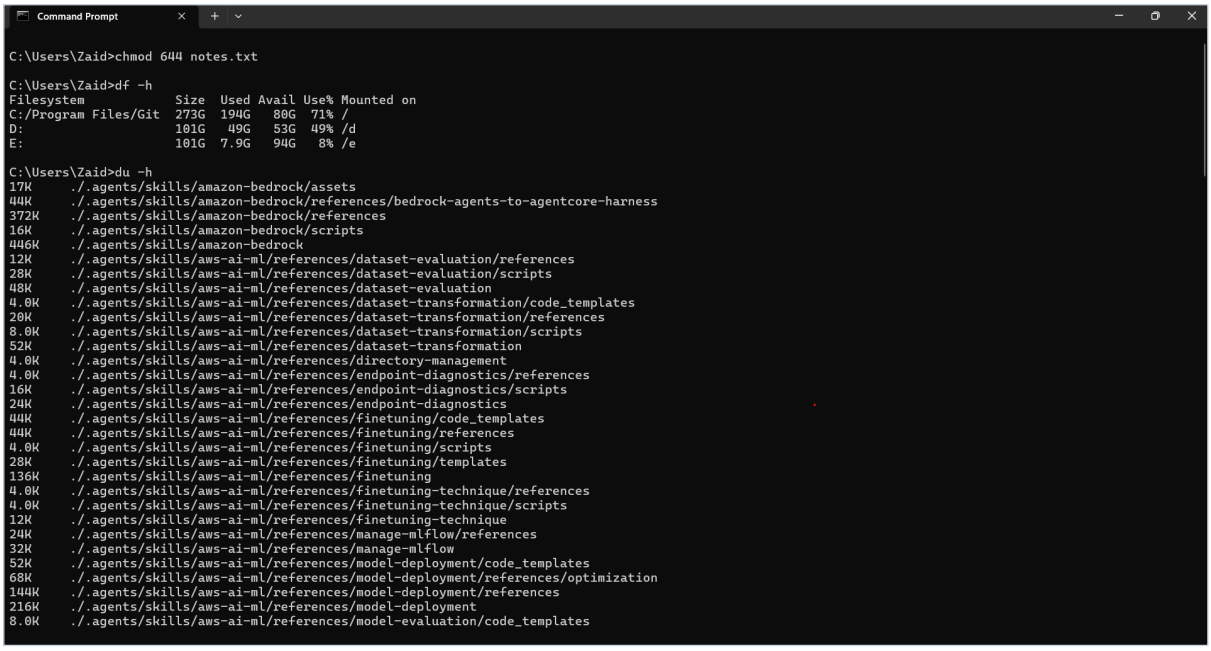
Terminal output 1: basic commands (pwd, whoami, date, uname -a, hostname, ls -lah)



Terminal output 2: directory and file management (mkdir, ls, cd, touch, echo, cat, cp, mv, rm)

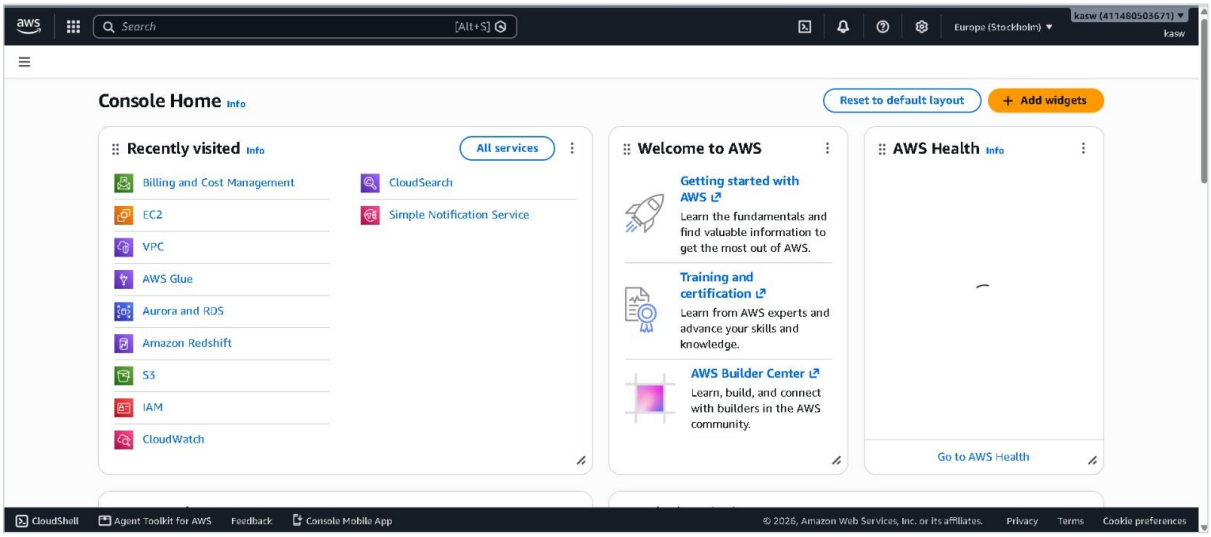


Terminal output 3: permissions and disk usage (chmod 644, df -h, du -h)



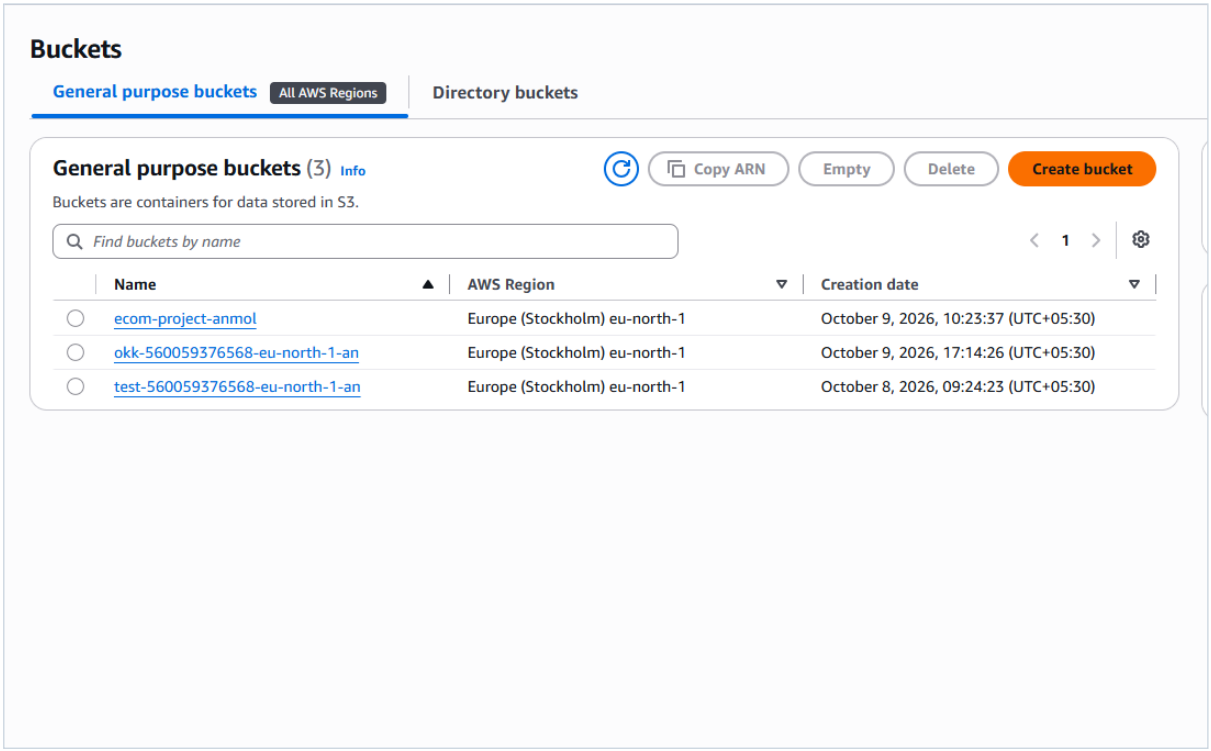
Account and Region Overview

AWS Console Home opened in the Europe (Stockholm) region

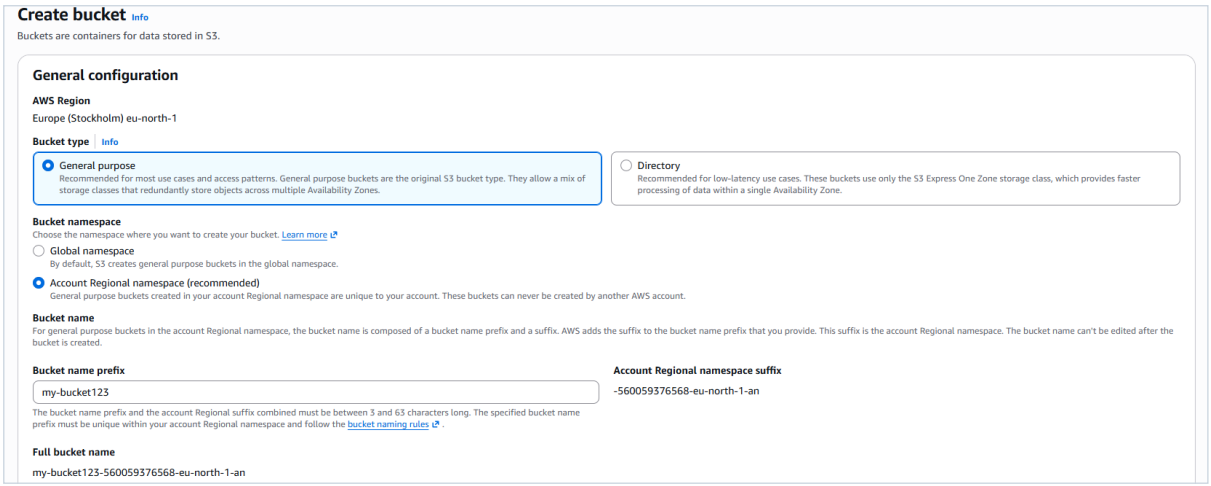


S3 Storage Setup

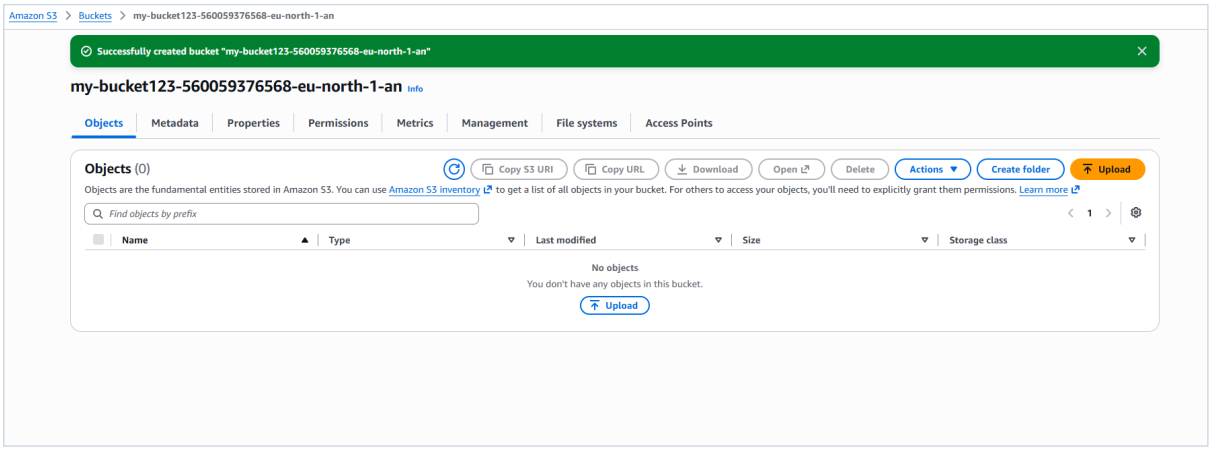
S3 buckets page in the Europe (Stockholm) eu-north-1 region



Creating a general purpose bucket in eu-north-1 named my-bucket123-560059376568-eu-north-1-an

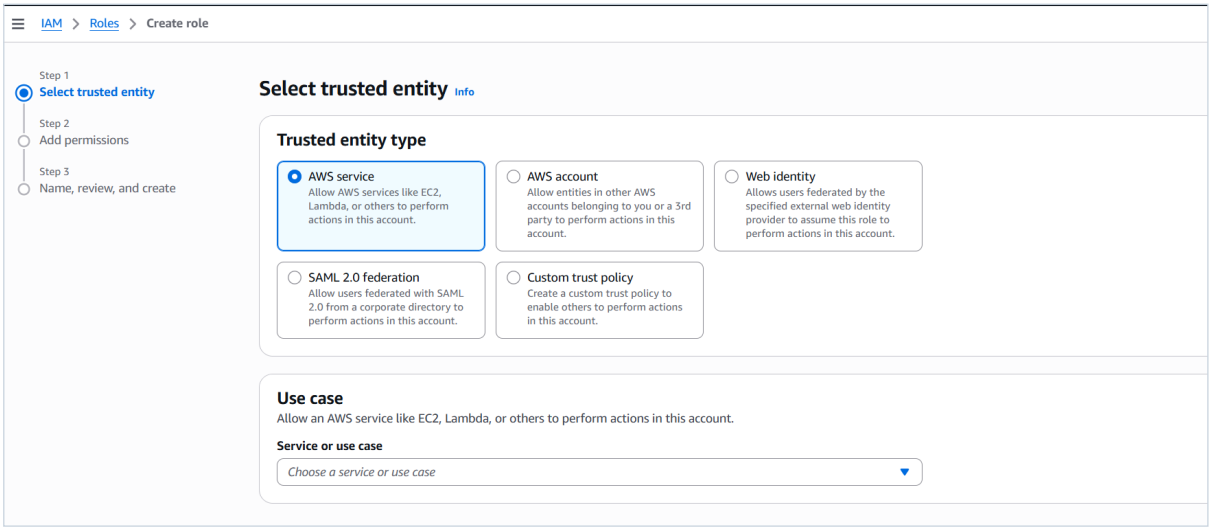


New bucket created successfully; it holds 0 objects so far

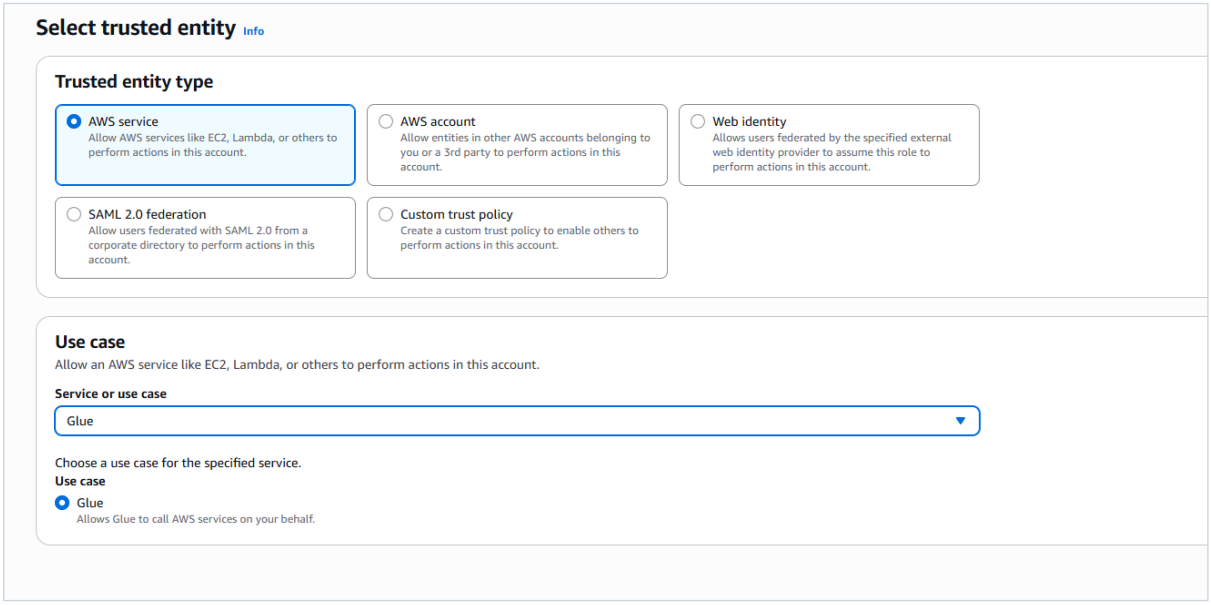


IAM Role Creation

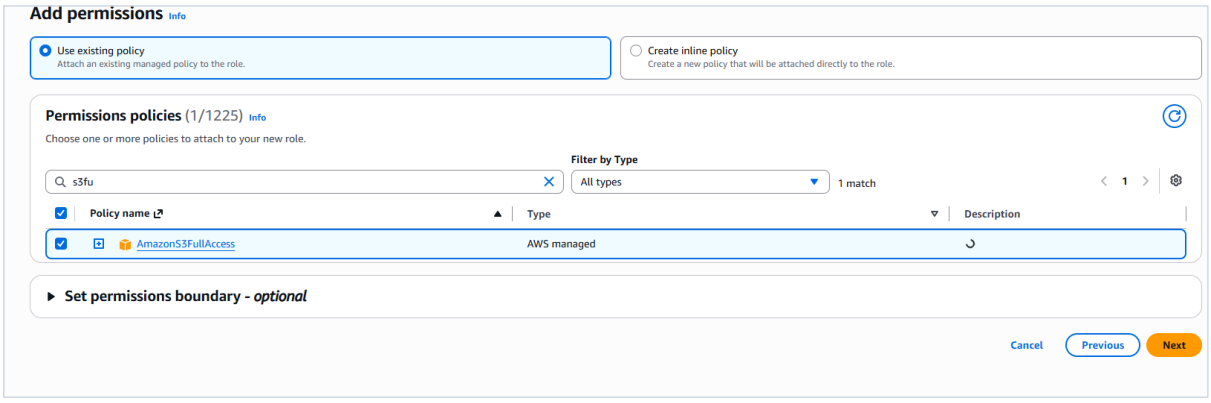
Trusted entity type chosen: AWS service



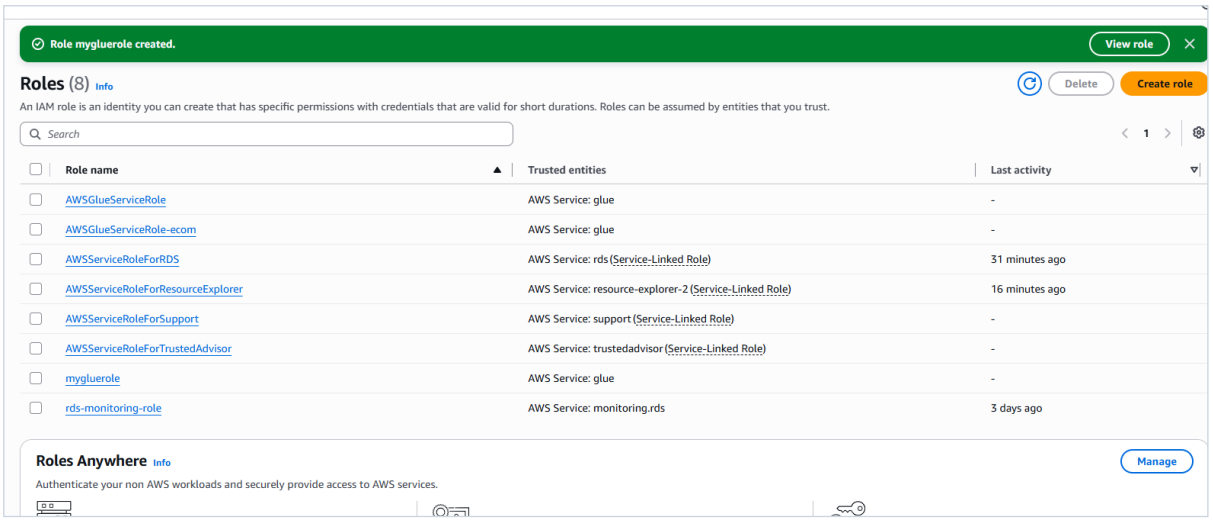
Glue chosen as the service use case



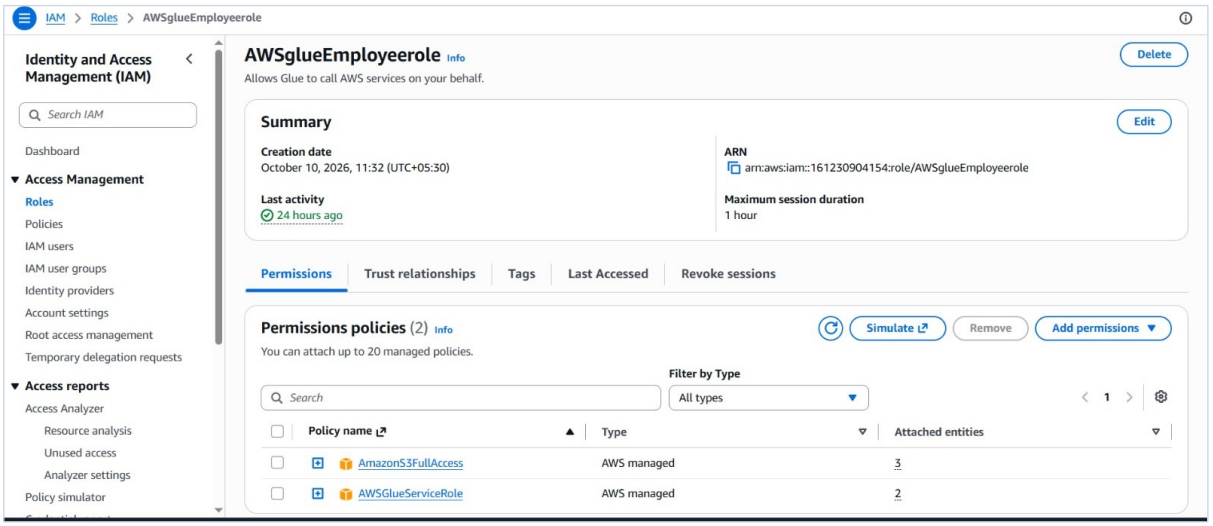
Permissions step with the AmazonS3FullAccess policy ticked



Confirmation message that the role mygluerole was created

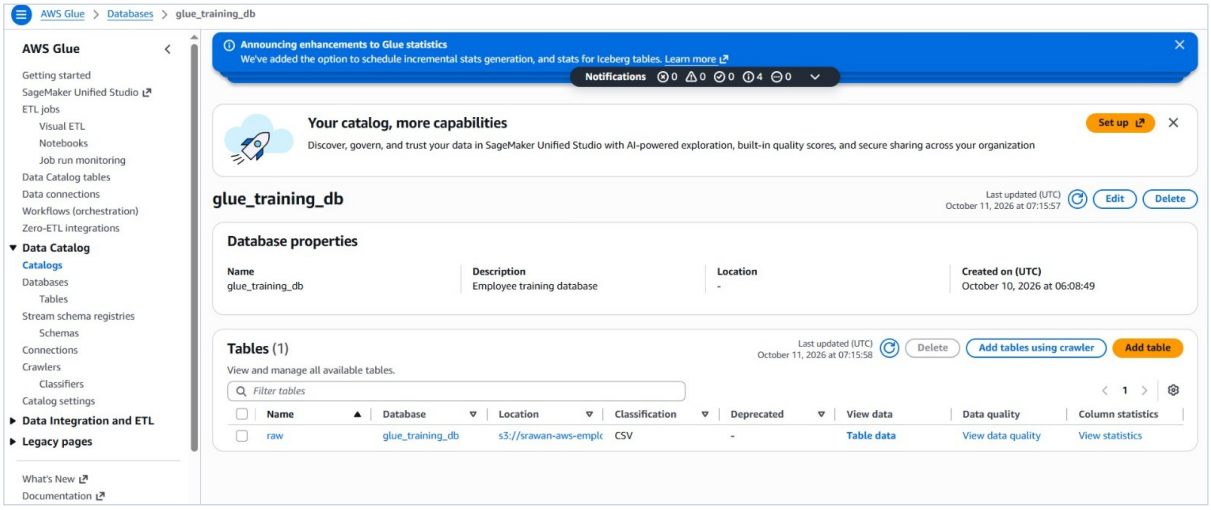


Role used by AWS Glue: AWSglueEmployeerole, carrying the AmazonS3FullAccess and AWSGlueServiceRole policies



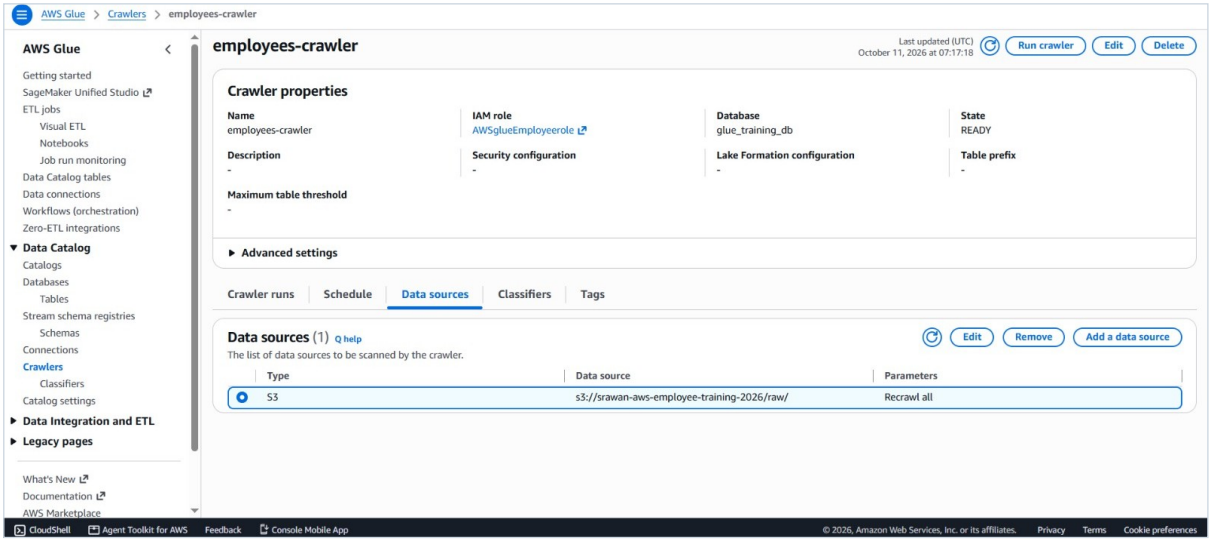
Glue Database Setup

Data Catalog database glue_training_db showing its properties and table list

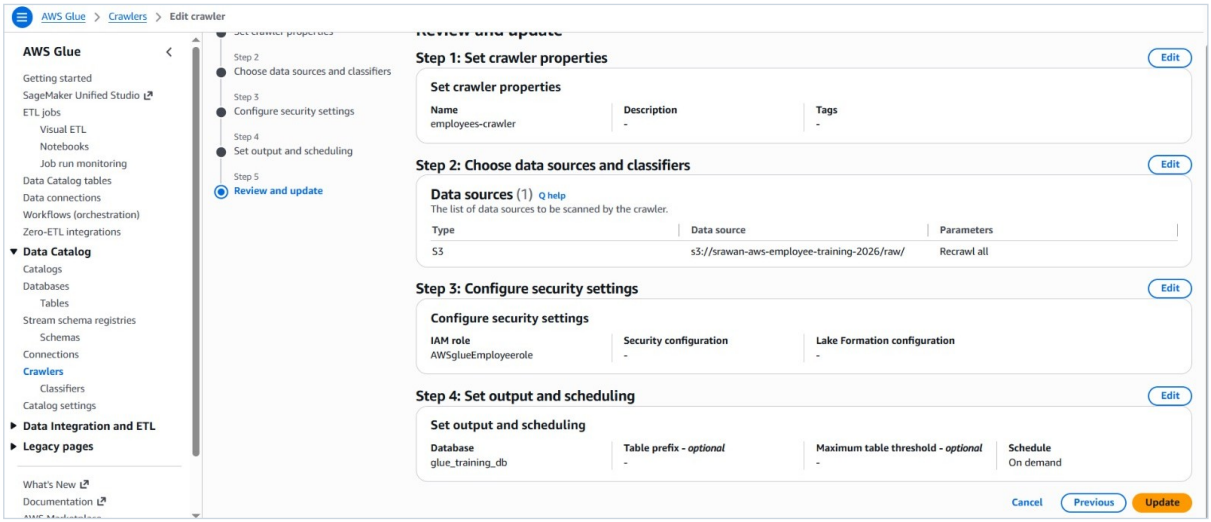


Crawler Setup

Crawler employees-crawler with an S3 source (raw/ folder), IAM role AWSglueEmployeerole and target database glue_training_db

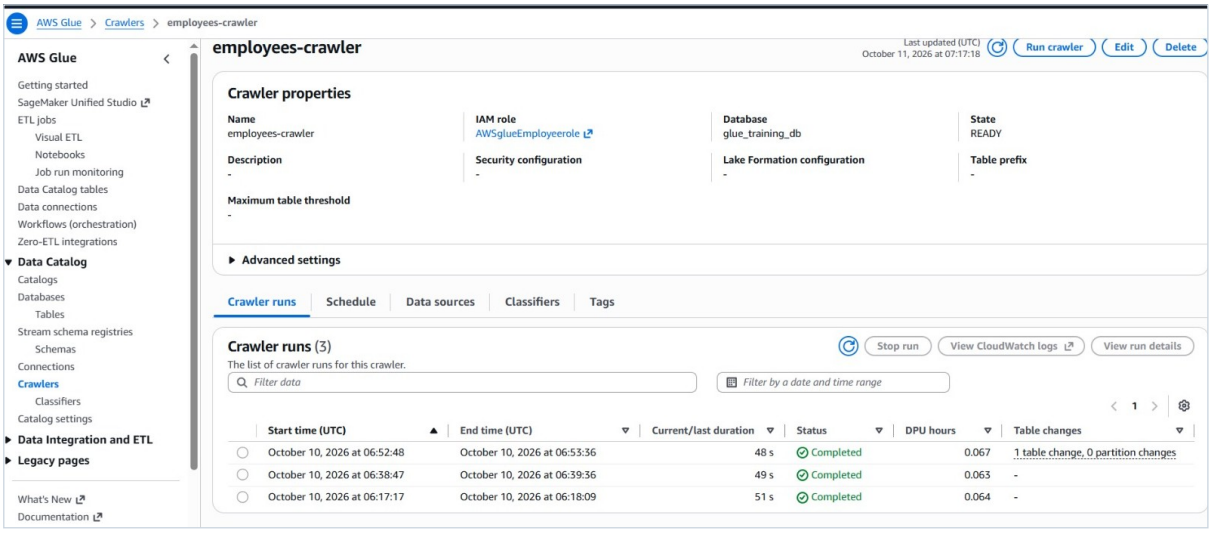


Review page of the crawler: properties, data source, security settings and output, with an on-demand schedule



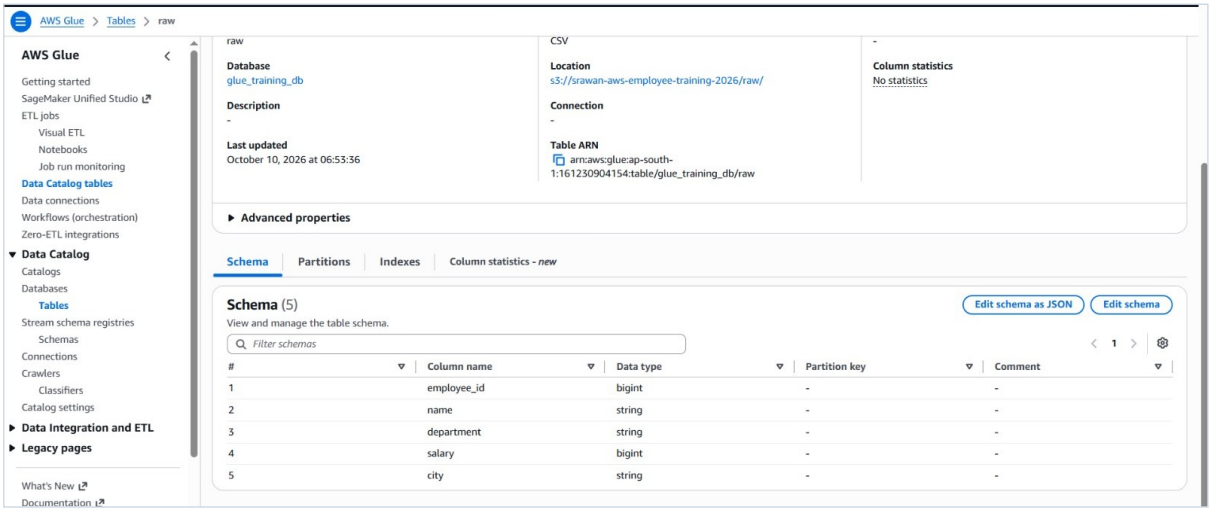
Running the Crawler

Crawler runs listed as Completed, with the crawler state showing READY



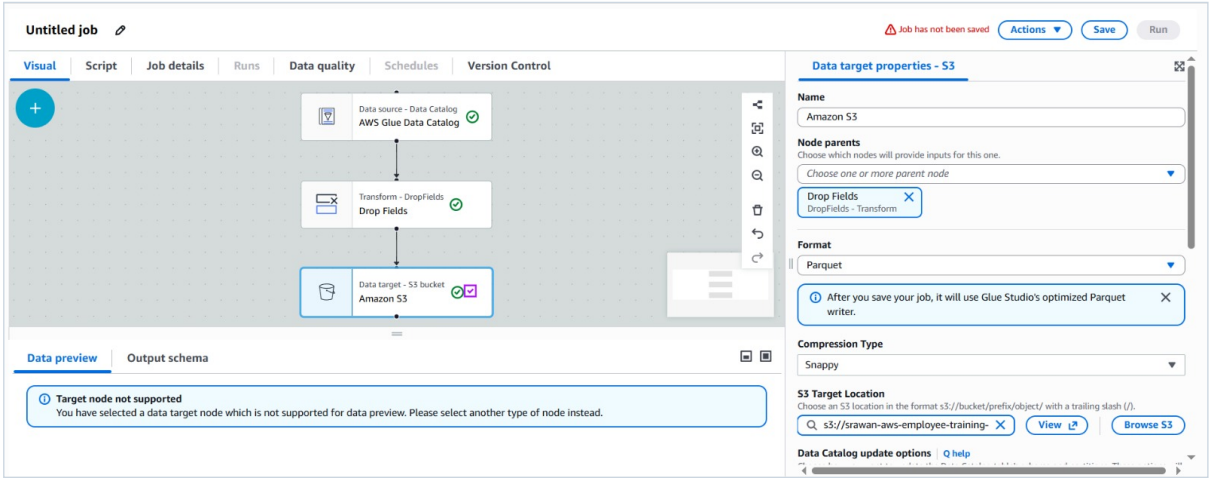
Catalog Table Details

Table raw generated by the crawler in glue_training_db, along with its schema and column types



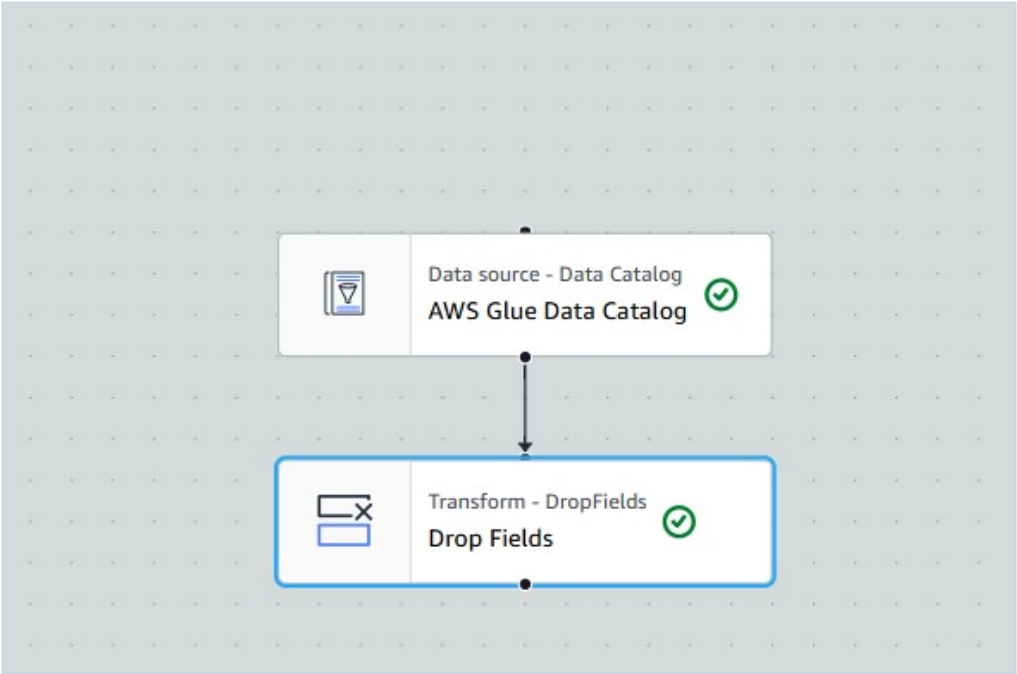
Visual ETL Job Design

Visual job with a Data Catalog source, a Drop Fields transform and an Amazon S3 target



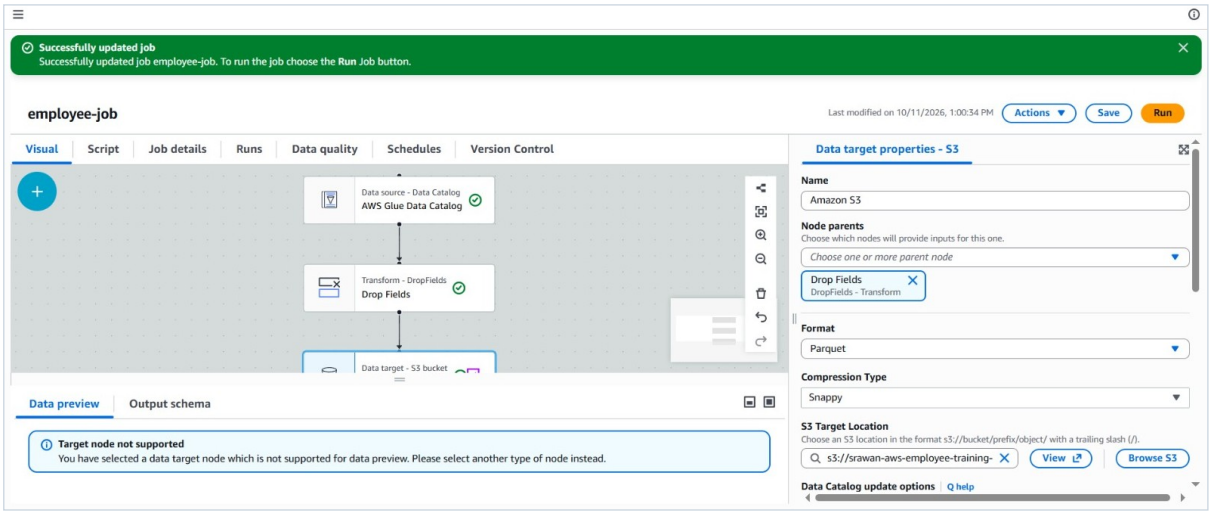
Data Transformation Step

Drop Fields node linked to the AWS Glue Data Catalog source



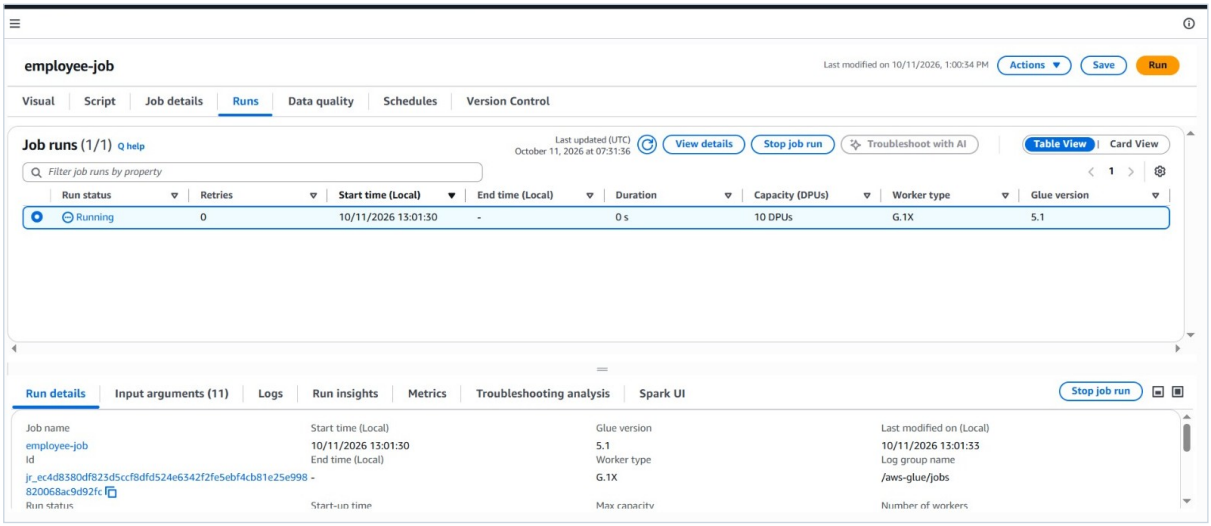
Output Target Settings

S3 target set to Parquet format with Snappy compression, plus the target location

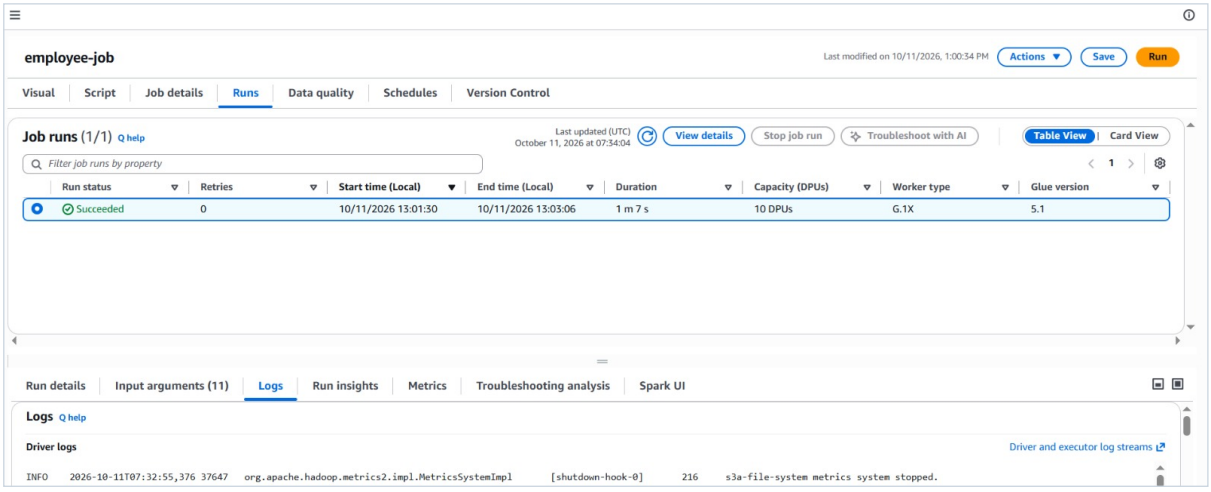


Running the ETL Job

employee-job triggered; the run status shows Running

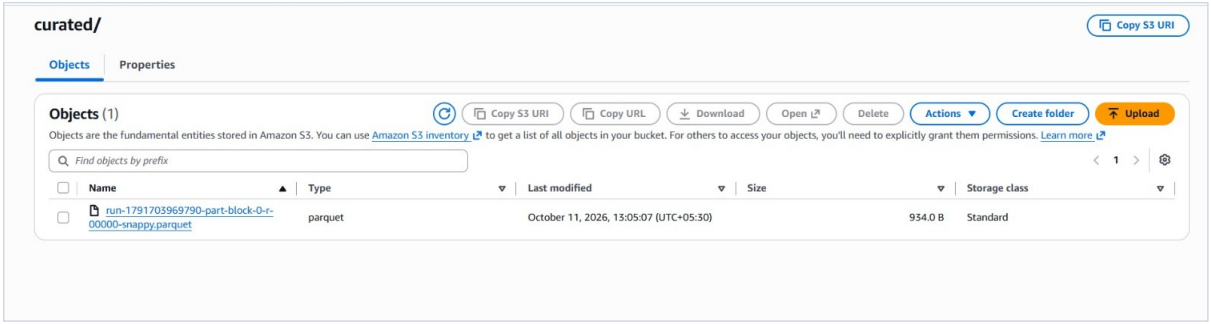


employee-job finished; the run status shows Succeeded



Curated Output in S3

Parquet file written by the job into the curated/ folder



Final Explanation

Overall idea

I took a raw employee CSV from S3, let Glue understand it, dropped one column, and saved it back as Parquet.

1. Amazon S3

I made a bucket with two folders: raw for the original CSV and curated for the cleaned output, so they never get mixed up.

2. IAM role

Glue cannot touch my bucket on its own, so I created a role with S3 access and the Glue service policy for it.

3. Glue crawler

I pointed a crawler at the raw folder; it read the file and worked out the columns and their types. All runs show Completed.

4. Data Catalog

The crawler saved those details as a table called raw in glue_training_db, with employee_id, name, department, salary and city.

5. Visual ETL

I built employee-job by joining boxes: read the table, change the data, write to S3. No coding needed.

6. Transformation

I used Drop Fields to remove the city column since I did not need it, and the rest passed through unchanged.

7. Output

The job wrote a Parquet file with Snappy compression into curated. It went from Running to Succeeded, and the file was there.

Workflow: S3 (raw) → Glue Crawler → Glue Data Catalog → Glue Visual ETL → Drop Fields → S3 (curated, Parquet)